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Epistasis and the release of genetic variation during long-term selection

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It is an enigma how long-term selection in model organisms and agricultural species can lead to marked phenotypic changes without exhausting genetic variation for the selected trait. Here, we show that the genetic architecture of an apparently major locus for growth in chicken dissects into a genetic network of four interacting loci. The interactions in this radial network mediate a considerably larger selection response than predicted by a single-locus model.

Long-term selection experiments have for decades provided insights into the genetic and biological mechanisms underlying complex quantitative traits[1,2]. Mutational variation has been shown to contribute to long-term selection response[3], but despite a wide range of empirical observations suggesting the importance of genetic interactions for selection response[4], there is a dearth of evidence on the biological and genetic mechanisms underlying these effects. The contribution of individual loci to selection response and evolutionary diversification has been explored using quantitative trait loci (QTL) mapping in model systems such as crosses between divergent populations[5] or selected lines derived from the same base population[6,7]. Some reports suggest that phenotypic changes are caused by few factors of large effects[6] or many factors of small effects[2] without any contribution of interlocus interactions (epistasis). Others report involvement of many loci with a significant contribution of epistasis[5,8,9]. A more in-depth understanding of the genome-wide mechanisms involved in selection responses, release of standing genetic variation and long-term evolutionary change necessitates explorations beyond the statistical estimates of these QTL effects.

The role of interactions in the genetic architecture of complex traits can be explored by epistatic QTL mapping[10]. These studies require large mapping populations, as the number of considered genotype classes and statistical tests increases exponentially with the number of loci. For example, a standard F_2 QTL analysis models three genotypes and $\sim 10^3$ genomic locations, whereas a two-locus epistatic analysis models nine genotype classes and $\sim 10^6$ two-locus combinations. The population size ultimately dictates the types of genetic interactions that can be identified, and we recommend at least a fourfold increase of the population size in epistatic QTL mapping to facilitate detection of QTL that express their effects in only one homozygote genotype class (that is, dominant epistasis). Here, we generated an intercross of 795 F_2 individuals between two divergent chicken lines obtained by bidirectional selection for body weight at 56 d of age[4] (**Fig. 1a**); the base population consisted of a cross of seven partially inbred lines. The mean body weight at 56 d of age was 624 g (range: 134–1,179 g; **Fig. 1b**).

Genomic loci (QTL) that contributed to the line difference, either by themselves or by interacting with other loci, were mapped in this F_2 population using our method for detection of epistatic interactions. We used empirical significance thresholds, derived by randomization testing, to test the significance of individual loci and combinations of loci[8] (detailed description in **Supplementary Methods** online). We found one QTL on GGA7 around 45 cM with significant marginal effects on body weight at 56 d of age (additive effect, 0.4 σ_P; $P_{\text{genome-wide}} < 0.01$); this locus was denoted Growth9 in our previous standard QTL analysis[7]. Furthermore, we detected five QTL pairs with significant epistatic effects ($P_{\text{pair, genome-wide}} < 0.05$; $P_{\text{interaction}} < 0.01$; **Supplementary Note** online). All five epistatic pairs involved Growth9 (**Fig. 1c**). We explored the relationship between statistical epistasis and biological gene interactions by visualizing the genotype-phenotype map for each of the five significant epistatic QTL pairs (**Fig. 1d–f**; **Supplementary Fig. 1** online; locus designation as in ref. 7). This analysis showed that Growth4, Growth6 and Growth12 had a significantly larger effect on growth ($P = 0.01$–0.001) in homozygous high-line Growth9 individuals than in Growth9 low-line homozygous individuals (**Fig. 1d–f**; **Table 1**). The four-locus network accounts for 45% of the high-line superiority (or 3.3 F_2 phenotypic standard deviations (σ_P)) in a homozygous high-line Growth9 background (**Table 1**) with virtually no effect in a homozygous low-line Growth9 background. The mean and variance of the F_2 population is expected to change if the low-line alleles at the four QTL in the network were exchanged for high-line alleles (**Fig. 1b**).

Figure 1g illustrates the phenotypic effect of total high- and low-line homozygosity for Growth4, Growth6 and Growth12, conditional on Growth9 genotype. Low-line homozygosity at Growth9 overrides any growth-increasing effects from high-line homozygosity at Growth4, Growth6 and Growth12. Furthermore, the high-line Growth9 allele has no apparent growth effects in homozygous low-line background of Growth4, Growth6 and Growth12. Thus, the marginal additive effect of Growth9, which is detected in a standard QTL scan, is not caused by an individual genetic effect of Growth9 but rather by the combination of beneficial alleles at the four loci. The relationship between increasing body weight and increasing high-line homozygosity is nearly linear, and we can obtain a rough estimate of

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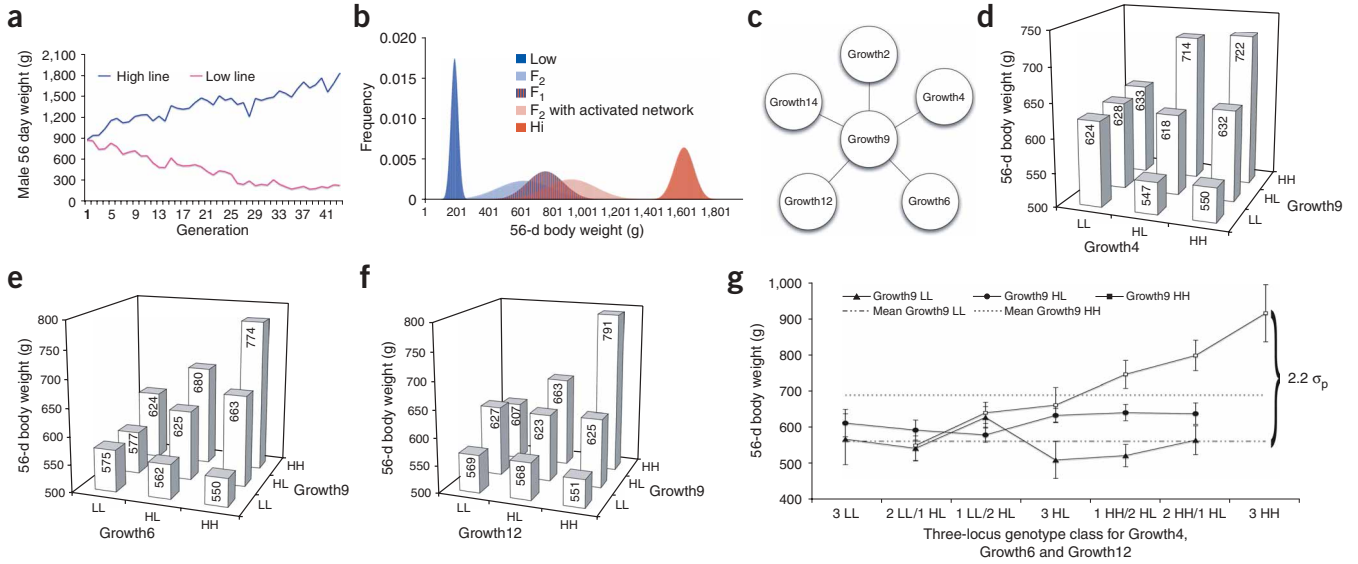


Figure 1 Radial network of interacting loci causing a marked amount of phenotypic change in a long-term chicken selection experiment. (**a**) Male body weight at 56 d of age in generations 1 to 42 for the high– and low–body weight lines. (**b**) Phenotypic distributions for high– and low–body weight lines, their F_1 and F_2 crosses and a fictive F_2 population in which the original F_2 mean and variance has been substituted for an expected mean and variance in a population fixed for high-line alleles at the four loci in the detected epistatic network but identical everywhere else in the genome. (**c**) A genomic network of five interacting loci underlying response to selection for body weight at 56 d of age. A line connecting the loci indicates a significant epistatic interaction ($P_{interaction} < 0.01$; **Supplementary Note**). GrowthX indicates a genome-wide significant epistatic QTL affecting body weight ($P_{pair,\ genome\text{-}wide} < 0.05$; **Supplementary Methods**), where X is the locus number as in ref. 7. (**d–f**) Phenotypic means in the nine possible two-locus genotype classes for three epistatic QTL pairs in which the growth-promoting effects of either locus is mainly activated in the high-line homozygous background for the other locus. There are no significant genetic effects in individuals homozygous for the low-line alleles at loci Growth4 (**d**), Growth6 (**e**), Growth9 (**d–f**) or Growth12 (**f**; **Table 1**). (**g**) Phenotypic means within alternative Growth9 genetic background plotted as a function of the degree of high- and low-line homozygosity at its interacting loci, Growth4, Growth6 and Growth12. The error bars represent s.e.m., estimated from more than three individuals.

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combined genetic effect of the four-locus network as a substitution effect of low-line homozygotes for Growth4, Growth6 and Growth12 in Growth9 high-line homozygote background. The difference between the two-locus double low-line homozygote and the three-locus triple high-line homozygote is 2.2 σ_P (or 38% of the phenotypic range in the F_2 population), which is close to the multiple regression estimate of 3.3 σ_P (or 45% of the parental line difference) in **Table 1**.

In our F_2 population, the high-line alleles were at an intermediate frequency, and consequently the number of individuals with advantageous combinations of alleles at the four interacting loci was low. As a result of this, only one of the six significant epistatic loci had significant marginal effects on BW56 (Growth9; $P_{genome\text{-}wide} < 0.01$). Four of the loci, however, had suggestive effects on BW56 (Growth2 and Growth6) or suggestive (Growth4) or significant effects (Growth12; $P_{genome\text{-}wide} < 0.01$) on other correlated traits[7]. In contrast, if four loci had individual additive effects that mediated the same increase in growth, each would have an effect of about 0.8 σ_P (100% larger than Growth9) and would be highly significant in a genome scan.

Thus, conventional QTL analysis uncovered only a single locus, Growth9, that had a minor effect (0.4 σ_P); however, our pairwise approach to test for locus interactions demonstrated that Growth9 lay at the center of a radial network in which pairwise interactions between Growth9 and each of three additional loci accounted for a marked amount of phenotypic change. The detected genetic network with multiple synergistic epistatic interactions provides a potential mechanistic explanation for the ability of populations to respond so markedly to artificial or natural selection. The marginal effect of alleles in the genomic network will increase as the directional selection for increased growth progresses. As the frequencies of the growth-promoting high-line alleles gradually increase, more individuals acquire beneficial allelic combinations, which activates the genetic network and releases standing genetic variation in response to selection. In this way, the selection response from this type of genomic network will mimic that of a genetic architecture involving many loci of small effects. Notably, these results provide the first empirical support for a recently proposed theoretical model for response to selection called Selection-Induced Genetic Variation (SIGV)[11].

Table 1 Genetic effects of four epistatic QTLs for body weight in chicken dependent on the genetic background

		Estimated additive effect (± s.e.m.)	
Tested QTL	Background QTL	High-line	Low-line
Growth4	Growth9	42.7 ± 23.3[a,c]	–43.7 ± 27.2
Growth6	Growth9	97.7 ± 31.8[a,d]	–8.2 ± 33.9
Growth9	Growth4	90.3 ± 20.3[b,d]	–4.1 ± 21.3
Growth9	Growth6	113.5 ± 23.9[a,d]	38.4[c] ± 21.9
Growth9	Growth12	118.7 ± 21.0[b,d]	18.6 ± 19.2
Growth12	Growth9	93.5 ± 21.1[b,d]	–10.6 ± 18.9
Sum A:		556.4	–9.6
Percentage population difference:		44.8	0.8

Estimated additive effect and s.e.m. for the tested QTL using the subset of F_2 animals homozygous for the high-line or low-line allele at the background QTL. [a]$P \leq 0.01$, [b]$P \leq 0.001$ in a one-sided *t*-test for a difference in the effects on growth of the locus in high-line and low-line homozygote backgrounds. [c]$P < 0.05$, [d]$P \leq 0.001$ in a one-sided *t*-test for a significant growth-increasing genetic effect of the locus in high-line homozygote background.

The type of interaction found between the loci in this population resembles classical mendelian epistatic segregation patterns, including duplicate epistasis and complementary epistasis. The ability of the loci in our network to release or suppress genetic variation is also related to the mechanisms shown by capacitors of cryptic genetic variation that have been described earlier in situations resulting from induced environmental or genetic perturbations. Such is the case for heat-shock protein 90 (*Hsp90*) in morphogenetic response to environmental cues in *Drosophila melanogaster*[12] and *Arabidopsis thaliana* and the epidermal growth factor receptor (*Egfr*) in photoreceptor determination in *D. melanogaster*[13]. Networks of capacitors have been described as an important component of the regulation of olfactory behavior in *D. melanogaster*[14]. The finding of a similar mechanism in a genetic network explaining the genetic architecture of long-term selection response in a segregating population could have important consequences for how to approach genetic dissection of complex traits as well as studies of the genetic mechanisms underlying evolutionary change. The observation that synergistic epistatic interactions are an integral part of a radial genetic network of interacting loci also highlights the relevance of gene interactions in the genetic architecture of complex traits and supports theoretical and empirical work in model systems that already indicates an important role of gene interaction networks in trait regulation[14,15].

Some insights to the biological function of the network can be gained by analyses of data on other traits collected in this cross (**Supplementary Note**). Further functional dissection at the molecular level is not possible until the underlying genes have been identified. Our findings, however, indicate that there might be a functional connection between theoretical models on long-term selection in quantitative genetics and the literature and experimental research on phenotypic robustness in model organisms. The results advance our understanding of the genetic architecture of metric traits, and they are relevant for the selective breeding of domestic animals and crops and for genetic studies of evolution and adaptation.

Note: Supplementary information is available on the Nature Genetics website.

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COMPETING INTERESTS STATEMENT

The authors declare that they have no competing financial interests.

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